

STADIOcare Group – Data Request

Project focus: Predicting patient demand, bed utilisation and length-of-stay patterns to support proactive capacity and workforce planning.

The request is intentionally detailed so that data quality, joinability, missing fields and historical coverage can be assessed before modelling begins. Please provide the most granular historical records available, preferably at patient/event level rather than aggregated monthly summaries. Use a consistent de-identified patient identifier where records need to be linked across systems.

1. Admissions, Discharges & Patient Movements (ADT)

Column name	Data type	Example	Additional information
Patient ID	String, identifier	P00018452	De-identified persistent identifier; must link to the same patient across encounters where permitted.
Encounter ID	String, identifier	ENC20260814001	Unique identifier for each admission/encounter; no duplicates.
Hospital ID	String, categorical	H012	Stable code for each hospital; provide a separate hospital reference list if names are not included.
Province	String, categorical	GP	Use the nine province codes: GP, EC, FS, KZN, LP, MP, NC, NW, WC.
Admission datetime	Datetime	2026-08-14 09:42:00	Local date/time; use yyyy-MM-dd HH:mm:ss consistently.
Discharge datetime	Datetime	2026-08-18 13:10:00	Local date/time; must be after admission for completed encounters.
Admission source	String, categorical	Emergency	Document all valid categories, e.g. emergency, elective, transfer, referral.
Admission type	String, categorical	Inpatient	Provide the controlled list used by STADIOcare.
Ward	String, categorical	Orthopaedic Ward A	Ward/unit at admission and, where available, subsequent ward movements.
Procedure code	String, categorical	KNEE01	Standard internal or clinical procedure code; provide code dictionary.
Procedure description	String, categorical	Knee replacement	Description corresponding to procedure code.
Discharge status	String, categorical	Home	Controlled list; include transfer, death, rehabilitation/other where applicable.
Readmission flag	Integer, binary	1	1 = readmission within the defined 30-day window; 0 = otherwise. If already calculated, provide definition.
Length of stay	Numeric, decimal	4.14	Prefer calculated from timestamps in days; retain raw admission/discharge timestamps.
ICU/high-care flag	Integer, binary	0	1 = patient used ICU/high-care during encounter; 0 = no.
Bed ID	String, categorical	BED-ORTH-014	If available, useful for occupancy reconstruction.
Movement datetime	Datetime	2026-08-16 07:30:00	For each transfer between wards/beds, provide event-level movement records.
Movement type	String, categorical	Ward transfer	Controlled list such as admission, transfer, discharge.

2. Bed Capacity & Occupancy

Column name	Data type	Example	Additional information
Hospital ID	String, categorical	H012	Must match the ADT hospital identifier.
Ward	String, categorical	Medical Ward B	Stable ward/unit code.
Bed ID	String, categorical	BED-MED-022	Unique bed identifier where available.
Bed type	String, categorical	General	Include general, high-care, ICU and other relevant categories.
Available beds	Integer	42	Number of operational beds available for the reporting interval.
Occupied beds	Integer	36	Number occupied at the specified timestamp.

Column name	Data type	Example	Additional information
Out-of-service beds	Integer	3	Beds unavailable due to maintenance, isolation, staffing or other reason.
Occupancy rate	Numeric, decimal	0.857	If supplied, retain; calculate independently where possible. Prefer 0–1 or 0–100 consistently.
Timestamp	Datetime	2026-08-14 08:00:00	Use the finest available time interval; daily or hourly preferred.
Closure reason	String, categorical	Staffing	Reason bed was unavailable, if applicable.

3. Appointment Scheduling & Outpatient Demand

Column name	Data type	Example	Additional information
Appointment ID	String, identifier	APT900184	Unique appointment identifier.
Patient ID	String, identifier	P00018452	De-identified identifier; link where permitted.
Hospital ID	String, categorical	H012	Must match reference data.
Clinic/unit	String, categorical	Cardiology	Clinic or specialist service.
Booked datetime	Datetime	2026-08-01 10:15:00	When appointment was created.
Appointment datetime	Datetime	2026-08-20 14:00:00	Scheduled date/time.
Appointment type	String, categorical	Follow-up	Controlled list of appointment types.
Attendance flag	Integer, binary	0	1 = attended; 0 = did not attend.
Cancellation flag	Integer, binary	0	1 = cancelled.
Cancellation lead time	Numeric, decimal	2.5	Days between cancellation and scheduled appointment, if cancelled.
Lead time	Numeric, decimal	19	Days between booking and appointment.
Specialty	String, categorical	Orthopaedics	Specialty/service category.

4. HR Rostering, Hours, Overtime & Agency Usage

Column name	Data type	Example	Additional information
Employee ID	String, identifier	EMP004821	De-identified stable employee identifier.
Hospital ID	String, categorical	H012	Primary work location.
Role	String, categorical	Registered nurse	Include nursing and other relevant clinical/support roles.
Shift date	Date	2026-08-14	Date of scheduled/actual shift.
Shift start	Datetime	2026-08-14 07:00:00	Scheduled or actual start time; distinguish scheduled from actual.
Shift end	Datetime	2026-08-14 15:30:00	Scheduled or actual end time.
Scheduled hours	Numeric, decimal	8.0	Hours rostered.
Actual hours	Numeric, decimal	8.5	Hours worked.
Overtime hours	Numeric, decimal	0.5	Hours beyond ordinary rostered hours.
Agency flag	Integer, binary	1	1 = agency worker/shift; 0 = permanent/internal.
Agency cost	Numeric, decimal	1850.00	Currency amount; specify whether inclusive/exclusive of applicable costs.

Column name	Data type	Example	Additional information
Leave status	String, categorical	None	Controlled list of leave types.
Ward/unit	String, categorical	ICU	Work location for the shift.

5. Emergency / Rapid 1 Dispatch

Column name	Data type	Example	Additional information
Dispatch ID	String, identifier	R1-20260814-00182	Unique emergency call/dispatch identifier.
Call datetime	Datetime	2026-08-14 18:42:11	Timestamp when call was received.
GPS origin	String, coordinate	-26.1234,28.0456	Latitude/longitude; provide precision and coordinate reference system.
GPS destination	String, coordinate	-26.1750,28.0200	Destination coordinates where applicable.
Priority	String, categorical	High	Controlled priority classification.
Response start	Datetime	2026-08-14 18:44:02	Timestamp used to calculate response time.
Arrival datetime	Datetime	2026-08-14 18:56:21	Arrival at incident location.
Hospital arrival	Datetime	2026-08-14 19:18:42	Arrival at receiving facility where applicable.
Outcome	String, categorical	Transported	Controlled outcome list.
Receiving hospital	String, categorical	H012	Hospital receiving the patient, if applicable.
Response time	Numeric, decimal	14.17	Minutes; preferably calculated from raw timestamps.

6. Clinical & Case-Mix Variables

Column name	Data type	Example	Additional information
Patient ID	String, identifier	P00018452	De-identified stable identifier.
Encounter ID	String, identifier	ENC20260814001	Links clinical data to ADT encounter.
Age	Integer	67	Age at admission; avoid unnecessary exact birth date if age is sufficient.
Sex	String, categorical	F	Use the organisation's documented categories.
Diagnosis code	String, categorical	I10	Standard diagnosis coding system; provide code dictionary/version.
Comorbidity count	Integer	3	If available; also provide underlying coded comorbidities where permissible.
Clinical severity	Numeric/categorical	3	Use existing documented acuity/severity measure and its definition.
Procedure complexity	Numeric/categorical	2	Existing complexity measure, if available.
Discharge destination	String, categorical	Home	Controlled list.
30-day readmission	Integer, binary	1	1 = readmitted within 30 days; 0 = otherwise; include readmission encounter ID if available.

7. Reference Data & Data Dictionary

Column name	Data type	Example	Additional information
Hospital ID	String, identifier	H012	One row per hospital; must match all operational datasets.

Column name	Data type	Example	Additional information
Hospital name	String, categorical	STADIOcare Johannesburg	Provide official facility name.
Province	String, categorical	GP	Use standard province codes consistently.
Hospital type	String, categorical	Acute hospital	Controlled facility type.
Bed capacity	Integer	320	Licensed/registered and operational capacity should be distinguished.
Ward ID	String, identifier	WARD-014	Stable ward code.
Ward type	String, categorical	Medical	Controlled list.
Procedure code	String, categorical	KNEE01	Provide description, effective dates and coding system.
Diagnosis code	String, categorical	I10	Provide description, coding system and version.
Calendar date	Date	2026-08-14	Date dimension including day-of-week, month, quarter and public-holiday flag.

8. Data Quality, Lineage & Metadata

Column name	Data type	Example	Additional information
Source system	String, categorical	ADT	Name of originating system/table.
Record ID	String, identifier	ADT-8891021	Original source-system record identifier.
Extract datetime	Datetime	2026-09-20 09:00:00	When the record was extracted.
Record created datetime	Datetime	2026-08-14 09:42:11	Original creation timestamp where available.
Record updated datetime	Datetime	2026-08-14 09:45:22	Latest update timestamp where available.
Missingness reason	String, categorical	Not collected	Use where missing values have a documented reason; do not replace unknowns with zero.
Data quality flag	String, categorical	Validated	Existing source-system quality/status flag, if available.
Source version	String, categorical	ADT_v5	System/table extract or schema version.

Data Delivery and Technical Requirements

1. Historical period: provide the maximum usable history available, with priority given to the full periods identified in the briefing pack (ADT and billing: 5 years; appointments: 3 years; Rapid 1: 4 years; radiology: 6 years; HR: 4 years).
2. Granularity: provide event-level records wherever possible. Do not aggregate to monthly or annual summaries unless the underlying records are unavailable.
3. Identifiers: provide stable de-identified Patient ID and Encounter ID fields so that admissions, clinical, appointment and outcome records can be linked. Do not provide names, ID numbers, addresses or other direct identifiers unless specifically approved.
4. Time: retain raw timestamps and state the timezone. Do not round timestamps before delivery.
5. Code lists: provide a data dictionary for every categorical/code field, including valid values, definitions, coding systems and version/effective dates.
6. Missing data: retain null/missing values and provide a reason or flag where the source system records why a value is missing. Do not substitute zero for missing values.
7. Data quality: provide known data-quality issues, system changes, historical gaps, duplicated records and periods where definitions changed.
8. Join keys: document how Hospital ID, Ward ID, Patient ID, Encounter ID and other keys relate across systems.
9. Privacy: supply only the minimum information necessary for the project and de-identify patient and employee information before delivery, subject to STADIOcare governance and applicable requirements.
10. File format: CSV or Parquet is preferred for analytical datasets; provide separate reference/data-dictionary files. Preserve field names and data types consistently across extracts.
11. Validation sample: provide a small representative extract from each source before the full delivery so that schema, joins, data types and missingness can be tested early.
12. Target/label definition: agree upfront on the exact definitions of outcomes such as occupancy, high-demand period, length of stay and 30-day readmission before model development.

Why these data are required: The requested sources directly address the identified business problem: ADT data establishes demand, throughput and length of stay; bed data measures capacity and occupancy; appointment and Rapid 1 data capture demand signals; HR data links demand to staffing, overtime and agency costs; clinical and case-mix data helps explain variation in length of stay; and reference/metadata data enables reliable joins and interpretation. This gives the project enough detail to test whether historical patterns can support predictive capacity and workforce planning.